

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Name: G J D Sai Sharan
COURSE NAME: Deep Learning

Reg no: 192425152
COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 5

Duration: 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are:
H H T H H H T H H T
Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).
2. Out of 50 students, 42 passed an exam.
Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.
3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.
Find the MLE for the probability that a bulb is defective.
4. A biased die is rolled 60 times. The number 6 appears 18 times.
Estimate the probability of rolling a 6 using MLE.
5. Given:
Input (x) = 4
Weight (w) = 2
Actual Output = 10
Prediction:
 $y = w \times x$
 1. Find the predicted output.
 2. Calculate the error.
 3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

Code of Conduct:

I G J D Sai Sharan/192425152 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

G. J. D. Saq sharan

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

1. Maximum Likelihood Estimation (MLE) for Coin Toss

Question

A coin is tossed **10 times** and the outcomes are:

H H T H H H T H H T

Using Maximum Likelihood Estimation, estimate the probability of getting Heads (**p**).

Solution

Step 1: Given

- Total number of tosses (n) = 10
- Number of Heads (x) = 7
- Number of Tails = 3

Step 2: Likelihood Function

$$L(p) = p^x \times (1 - p)^{(n - x)}$$

Substitute the values:

$$L(p) = p^7 \times (1 - p)^3$$

Step 3: Log-Likelihood Function

$$\log L(p) = 7 \log(p) + 3 \log(1 - p)$$

Step 4: Differentiate

$$d(\log L)/dp = 7/p - 3/(1 - p)$$

Set the derivative equal to zero.

$$7/p = 3/(1 - p)$$

Step 5: Solve

$$7(1 - p) = 3p$$

$$7 - 7p = 3p$$

$$7 = 10p$$

$$p = 7/10$$

$$p = 0.7$$

Final Answer

The Maximum Likelihood Estimate (MLE) of the probability of getting Heads is:

$$p = 0.7 \text{ (70\%)}$$

2. MLE for Student Pass Probability

Question

Out of 50 students, 42 passed an exam.

Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.

Solution

Step 1: Given

- Total students (n) = 50
- Students passed (x) = 42
- Students failed = 8

Step 2: Likelihood Function

$$L(p) = p^{42} \times (1 - p)^8$$

Step 3: Log-Likelihood Function

$$\log L(p) = 42 \log(p) + 8 \log(1 - p)$$

Step 4: Differentiate

$$d(\log L)/dp = 42/p - 8/(1 - p)$$

Set the derivative equal to zero.

$$42/p = 8/(1 - p)$$

Step 5: Solve

$$42(1 - p) = 8p$$

$$42 - 42p = 8p$$

$$42 = 50p$$

$$p = 42/50$$

$$p = 0.84$$

Final Answer

The Maximum Likelihood Estimate (MLE) of the probability that a student passes is:

$$p = 0.84 \text{ (84\%)}$$

3. MLE for Defective Light Bulbs

Question

A factory produces light bulbs.

In a sample of 100 bulbs, 8 are defective.

Find the MLE for the probability that a bulb is defective.

Solution

Step 1: Given

- Total bulbs (n) = 100
- Defective bulbs (x) = 8
- Non-defective bulbs = 92

Step 2: Likelihood Function

$$L(p) = p^8 \times (1 - p)^{92}$$

Step 3: Log-Likelihood Function

$$\log L(p) = 8 \log(p) + 92 \log(1 - p)$$

Step 4: Differentiate

$$d(\log L)/dp = 8/p - 92/(1 - p)$$

Set the derivative equal to zero.

$$8/p = 92/(1 - p)$$

Step 5: Solve

$$8(1 - p) = 92p$$

$$8 - 8p = 92p$$

$$8 = 100p$$

$$p = 8/100$$

$$p = 0.08$$

Final Answer

The Maximum Likelihood Estimate (MLE) of the defective probability is:

$$p = 0.08 \text{ (8\%)}$$

4. MLE for Probability of Rolling a 6

Question

A biased die is rolled 60 times.

The number 6 appears 18 times.

Estimate the probability of rolling a 6 using MLE.

Solution

Step 1: Given

- Total rolls (n) = 60
- Number of sixes (x) = 18
- Other outcomes = 42

Step 2: Likelihood Function

$$L(p) = p^{18} \times (1 - p)^{42}$$

Step 3: Log-Likelihood Function

$$\log L(p) = 18 \log(p) + 42 \log(1 - p)$$

Step 4: Differentiate

$$d(\log L)/dp = 18/p - 42/(1 - p)$$

Set the derivative equal to zero.

$$18/p = 42/(1 - p)$$

Step 5: Solve

$$18(1 - p) = 42p$$

$$18 - 18p = 42p$$

$$18 = 60p$$

$$p = 18/60$$

$$p = 0.30$$

Final Answer

The Maximum Likelihood Estimate (MLE) of rolling a 6 is:

$$p = 0.30 \text{ (30\%)}$$

5. Prediction, Error, and Weight Update

Question

Given:

$$\text{Input (x)} = 4$$

$$\text{Weight (w)} = 2$$

$$\text{Actual Output} = 10$$

Prediction:

$$y = w \times x$$

Find:

1. Predicted output
2. Error
3. Updated weight if the gradient is 4 and the learning rate is 0.01

Solution

Step 1: Calculate Predicted Output

Prediction formula:

$$y = w \times x$$

Substitute the values:

$$y = 2 \times 4$$

$$\text{Predicted Output} = 8$$

Step 2: Calculate Error

$$\text{Error} = \text{Actual Output} - \text{Predicted Output}$$

$$\text{Error} = 10 - 8$$

$$\text{Error} = 2$$

Step 3: Update the Weight

Gradient Descent Formula:

$$\text{New Weight} = \text{Old Weight} - (\text{Learning Rate} \times \text{Gradient})$$

Substitute the values:

$$\text{New Weight} = 2 - (0.01 \times 4)$$

$$\text{New Weight} = 2 - 0.04$$

$$\text{New Weight} = 1.96$$

Final Answer

- Predicted Output = **8**
- Error = **2**
- Updated Weight = **1.96**